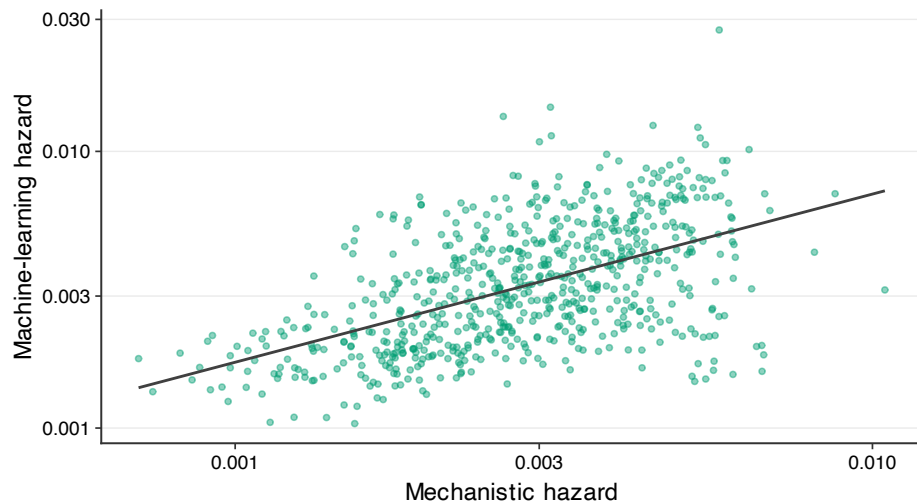


a

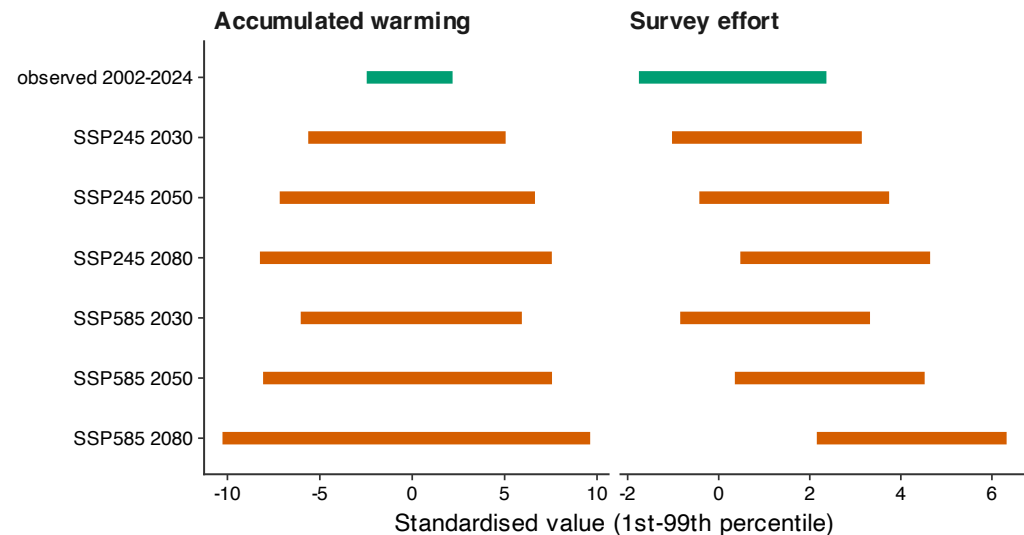
Within the observed range they track each other

Province-year mean predicted hazard, 2002-2024; moderate but clearly positive

**b**

Why: the scenarios leave the training range

Tree ensembles return boundary leaf values once covariates exit the observed support

**c**

Outside it they diverge completely

The mechanistic model projects increases in 176 of 186 province-scenario cells, the ML model in only 42

